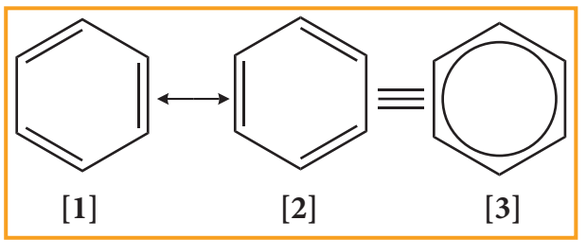
of atoms is called resonance. The actual structure of the molecule is said to be resonance hybrid of various possible alternative structures. In benzene, Kekule's structures I & II represented the resonance structure, and structure III is the resonance hybrid of structure I &II



The structures 1 and 2 exist only in theory. The actual structure of benzene is the hybrid of two hypothetical resonance structures.

6. Spectrosscopic measurments:

Spectroscopic measurements show that benzene is planar and all of its carbon-carbon bonds are of equal length 1.40A°. This value lies between carbon-carbon single bond length 1.54A° and carbon-carbon double bond length 1.34A°.

7. Molecular orbital structre:

The structure of benzene is best described in terms of the molecular orbital theory. All the six carbon atoms of benzene are sp^2 hybridized. Six sp^2 hybrid orbitals of carbon linearly overlap with six 1s orbitals of hydrogen atoms to form six C - H sigma bonds. Overlap between the remaining sp^2 hybrid orbitals of carbon forms six C-C sigma bonds.

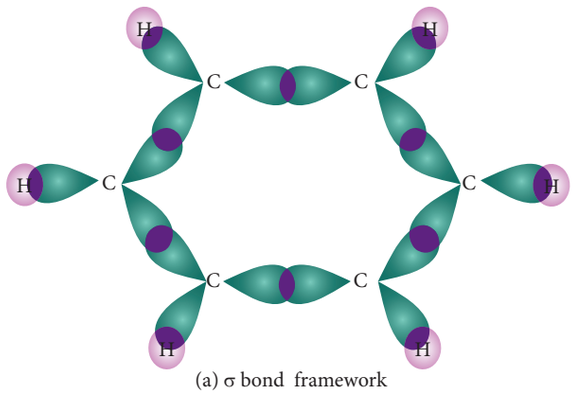


Figure13.6. Formation of Sigma bond in benzene

All the σ bonds in benzene lie in one plane with bond angle 120°. Each carbon atom in benzene possess an un hybridized p-orbital containing one electron. The lateral overlap of their p-orbital produces 3 π- bond The six electrons of the p-orbitals cover all the six carbon atoms and are said to be delocalised. Due to delocalization, strong π-bond is formed which makes the molecule stable. Hence unlike alkenes and alkynes benzene undergoes substitution reactions rather addition reactions under normal conditions.

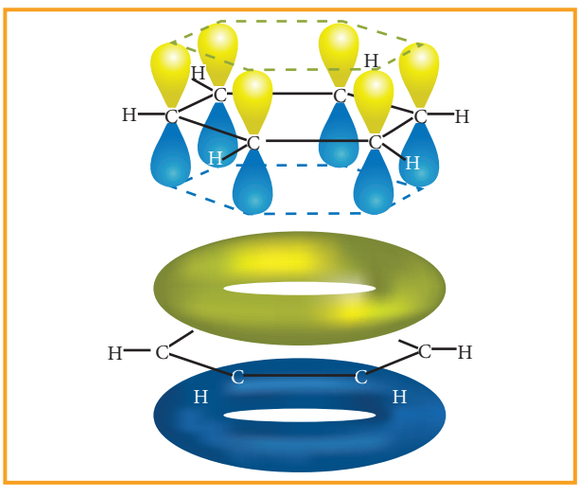


Figure 13.7. All carbon atoms have The delocalized π Mo is formed by

p orbitals the overlap of six p-orbitals

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